1 Appendix for Section 4

1.1 Growth Impatience Implies Harmenberg Impatience

{sec:AppxHarmImp

We show here that growth impatience implies the condition imposed by Harmenberg (2021) to guarantee the existence of a permanent income weighted distribution of normalized market resources. Letting f denote the density of the permanent income shock ψ, the impatience condition imposed by Harmenberg (2021) is

$$\log(\text{Þ}) < \int \log(\mathcal{G}\psi)\psi f(\psi)\, d\psi. \tag{1}$$ {eq:HarmImp}

Claim 1. *If growth impatience holds, then (1) holds.*

Proof. Since $\psi \mapsto \log(\mathcal{G}\psi)\psi$ is convex, Jensen's inequality implies

$$\int \log(\mathcal{G}\psi)\psi f(\psi)\, d\psi \geq \log(\mathcal{G}\mathbb{E}\psi)\mathbb{E}\psi = \log(\mathcal{G}) \tag{2}$$ {eq:Jensen}

Since growth impatience implies $\mathcal{G} > \text{Þ}$ and log is strictly monotone increasing, the result follows.

□

1.2 Apparent Balanced Growth in $\bar{c}$ and $\text{cov}(c, \boldsymbol{p})$

{sec:ApndxBalance

Section 4.2 demonstrates some propositions under the assumption that, when an economy satisfies the GIC, there will be constant growth factors $\Omega_{\bar{c}}$ and Ω_{cov} respectively for $\bar{c}$ (the average value of the consumption ratio) and $\text{cov}(c, \boldsymbol{p})$. In the case of a Szeidl-invariant economy, the main text shows that these are $\Omega_{\bar{c}} = 1$ and $\Omega_{\text{cov}} = \mathcal{G}$. If the economy is Harmenberg- but not Szeidl-invariant, no proof is offered that these growth factors will be constant.

1.3 $\log c$ and $\log \text{cov}(c, \boldsymbol{p})$ Grow Linearly

Figures 1 and 2 plot the results of simulations of an economy that satisfies Harmenberg- but not Szeidl-invariance with a population of 4 million agents over the last 1000 periods (of a 2000 period simulation).[1] The first figure shows that $\log \bar{c}$ increases apparently linearly. The second figure shows that $\log(-\text{cov}(c, \boldsymbol{p}))$ also increases apparently linearly. (These results are produced by the notebook `ApndxBalancedGrowthcNrmAndCov.ipynb`).

[1] For an exposition of our implementation of Harmenberg's method, see this supplemental appendix.

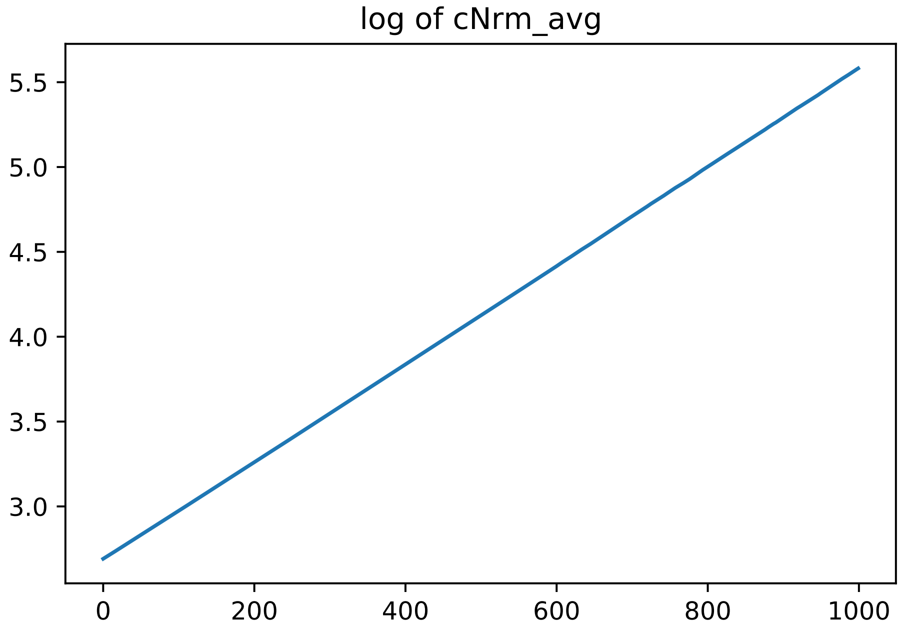

Figure 1 Appendix: log $\mathfrak{c}$ Appears to Grow Linearly {fig:logcNrm}

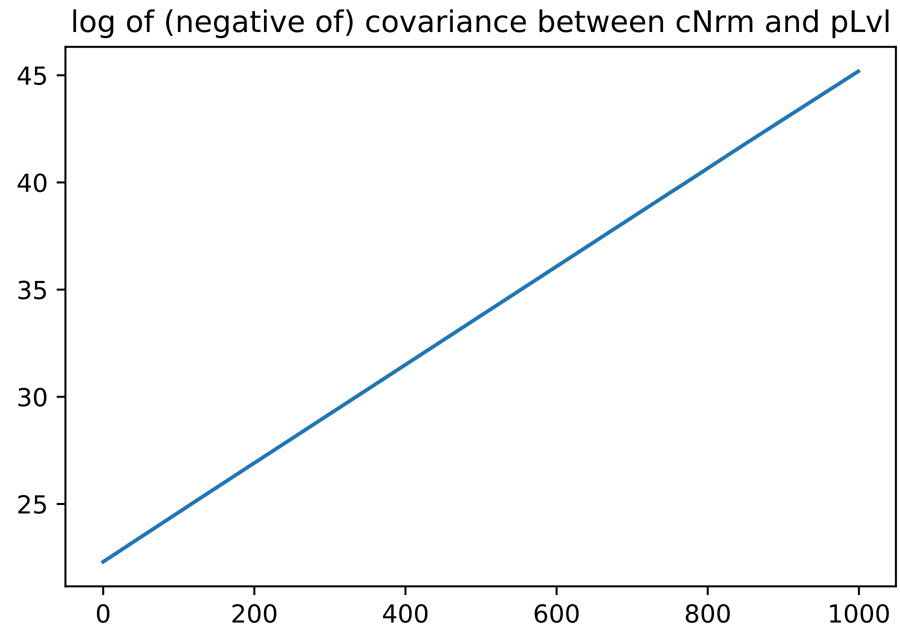

Figure 2 Appendix: log $(-\operatorname{cov}(c, \boldsymbol{p}))$ Appears to Grow Linearly {fig:logcov}

(see (2))

References

Harmenberg, Karl (2021): "Aggregating heterogeneous-agent models with permanent income shocks," Journal of Economic Dynamics and Control, 129, 104185.